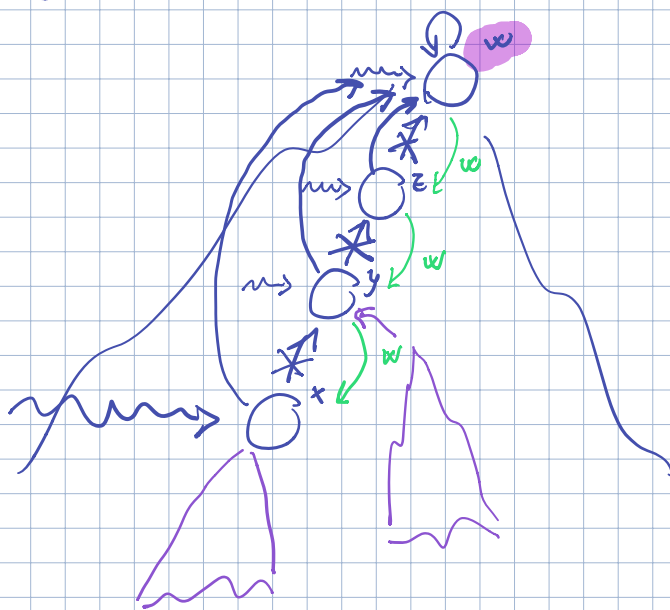


```

Find-PC(x) {
  if (x ≠ x.parent) {
    x.parent ← Find-PC(x.parent)
  } else {
    return x
  }
}

```



Comprimmo
solo il cammino
x-radice

Costi senza dim.

m operazioni HUF con Union by Rank
e P.C. di cui m MAKE hanno costo

$$! \quad O(m \cdot \underbrace{\alpha(m, m)}) \quad !$$

Inversa funz. Ackermann



e^n

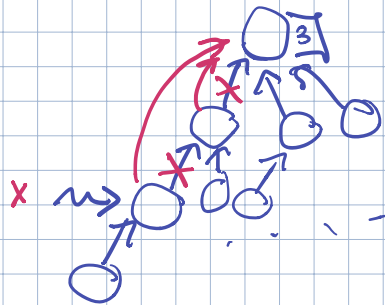
$\log n$

↓
Crescita lentissima
"Poco più di una
costante"

Nella migliore delle ipotesi avremmo
avuto comunque un costo almeno $\Omega(n)$

Rango

- Senza Path Compression $\leadsto$ Altezza dell'alb.
- Con Path Comp. $\leadsto$ Rango $\geq$ Altezza

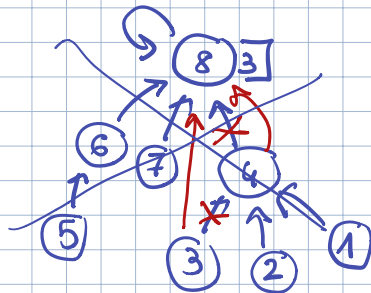
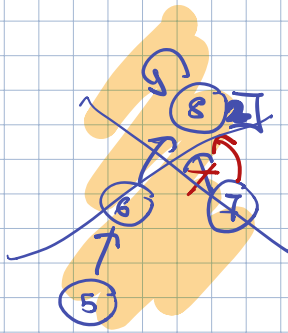
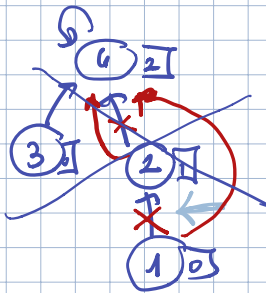
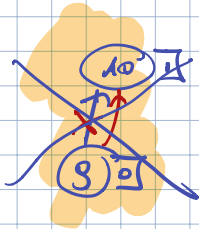
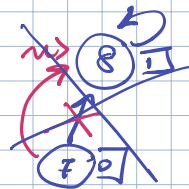
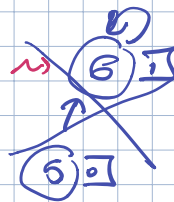
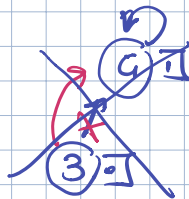
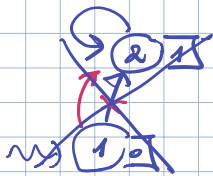


Approssimazione
per eccesso

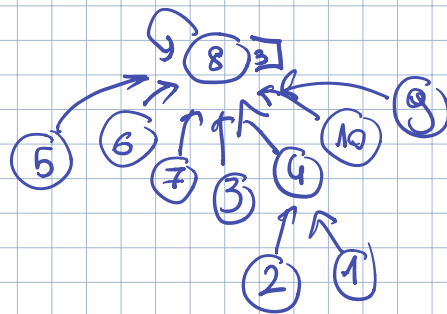
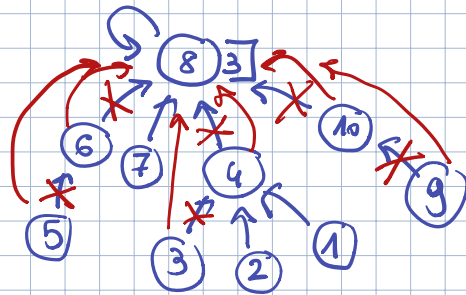
Es. 23/06/17

Alberi Union by rank PC (maggiore come rep)

1 10



Union(5,9)



~. ~. ~.

Costo Timeare

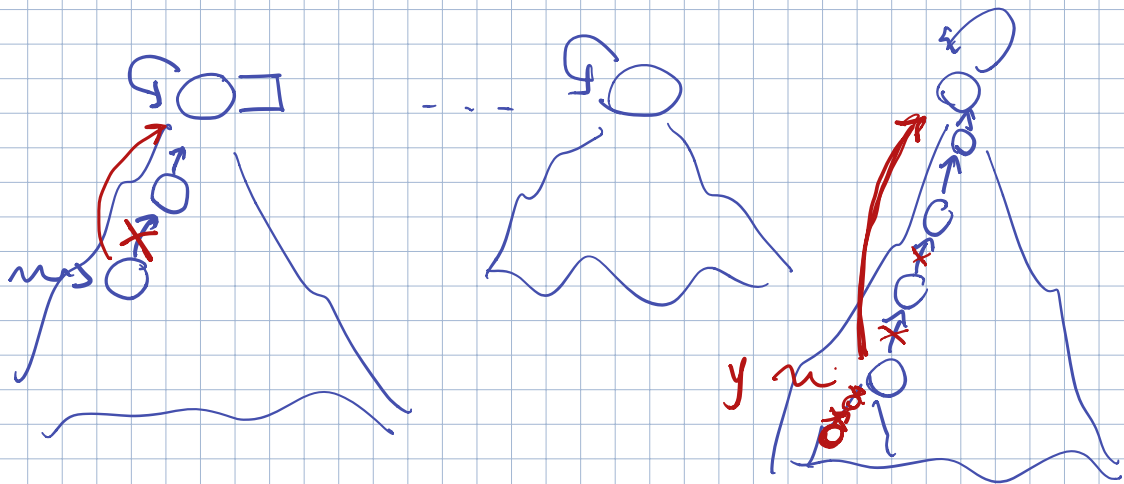
Union by Rank e PC $\leadsto O(m \alpha(m, m))$

Make $\leadsto$ Union $\leadsto$ Find
qui
zap.

Make $\rightarrow \mathcal{O}(1) \cdot m \rightarrow \mathcal{O}(m)$

Union $\rightarrow \mathcal{O}(1) \cdot m \rightarrow \mathcal{O}(m) \leftarrow$

Find $\mathcal{O}(m) \leftarrow$



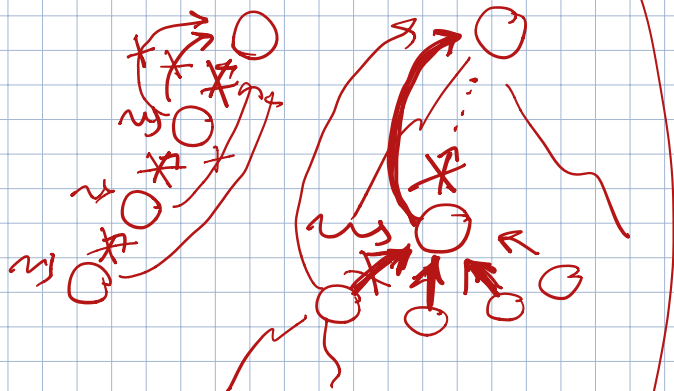
Ogni arco creato con Make o Union
Viene percorso al più una volta

$$O(m) \cdot O(1)$$

Conto il costo complessivo delle Find
contando quanti archi complessivamente
semplici durante le Find

Al più 1 volta
per archi
originali
 $O(m)$

Ogni arco nuovo
può essere percorso



• per Find
 del modo stesso
 • per una Find
 di uno dei Figli

La prima volta che eseguo Find(x)
 ho costo $O(\log n)$
 Tutte le successive Find(x) avranno costo
 $O(1)$

$$O(m \log n) + O(m - m) \cdot O(1)$$

$\uparrow$ $\uparrow$

$$O(m \log n + m) = m \cdot \Omega(\log n) \rightarrow O(m)$$

(Solo Union by rank $O(m \log n)$)

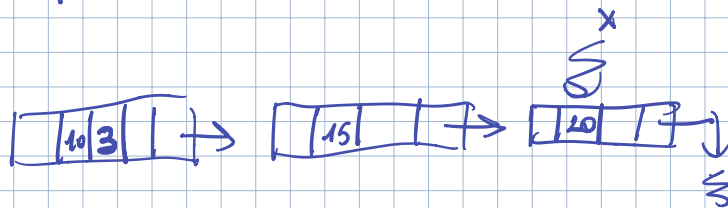
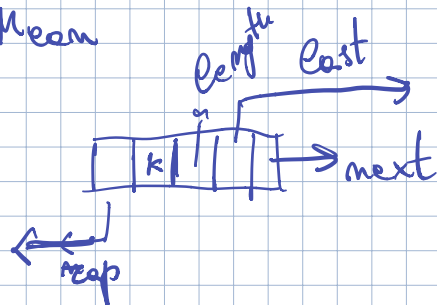
$$f \quad m = m + \underbrace{u}_{n-1} + (f) < \frac{n^2}{n^3}$$



Es. 18/06/18

Liste WU

Mean



Campo media aggiornato solo nel rep

Union come aggiornare la media?



nuova media

$$\frac{10 \cdot 10 + 5 \cdot 8}{18}$$

I costi non cambiamo

Oppure memorizzo la somma

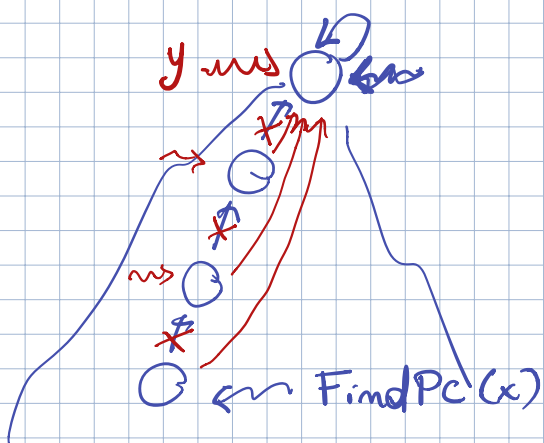
Union $Somma\ 1^a + somma\ 2^a$

Mean $\frac{somma}{lung.}$

Costi invarianti

~ ~ ~

22/01/2020 FindPC in Peace



FindPC_while(x) {

y ← x

while (y ≠ y.parent) {

y ← y.parent

}

while (x ≠ y) {

x ← x.parent

$O(\log n)$
1 op. Find

$$m = \textcircled{u}(m) \quad /$$

x. somme

Scrivere codice

22/07/18